

# Micro-VLA: Scaling In-Context Robotic Manipulation via Observation-Only Videos \*

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## 1 Chosen environment

We have chosen the Robosuite environment to work with. We have decided to use the related Robomimic dataset to train our models. The first task we wanted to achieve is to train a model on a task (e.g. Lift - which consists of picking up a red cube from the table), and evaluate it on a slight variation of the same task (e.g. changing the colour of the cube to blue, or changing the shape to a cylinder). The second task we wanted to achieve is to train a model on a set of tasks (e.g. Lift, and Nut Assembly - which consists of putting a nut on top of a bolt), and evaluate it on a more complex task (e.g. Stack - which consists of picking up a red cube from the table, and stacking it on top of a blue cube).

We had to postprocess the robomimic PH dataset as it wasn't publicly released in a full form with the images included and dense rewards.

We had also built an evaluation suite - an extendable framework for evaluating models on the Robosuite environment. It allows us to easily add new tasks, abstract away the implementation details of the environment, and of the models.

## 2 Initial Idea - finetuning a pretrained VLA model

Since the project had VLA in its name our initial idea was to start from a pretrained VLA model and to fine-tune it on our tasks. We have decided to use one called SmolVLA mainly because of its low computational requirements - most VLA would be out of our reach to fine tune on a single consumer grade (i.e. colab T4 with 14GB VRAM GPU). We have spent some time trying to get it to work - and have managed to get some first successful attempts (although very rare - success rate was approx. 10%) on the Lift task, but we have realized that with one epoch taking over 1h of compute this approach wasn't going to work well.

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\*we strayed from the observation only part

### 3 Pivot - traning a Decision Transformer from (almost) scratch

In order to be able to iterate faster and test more ideas we have decided to pivot and to train a Decision Transformer from scratch - but using a pretrained vision encoder (e.g. Dino). This combined with preprocessing the dataset to contain Dino embeddings of every frame allowed us to train a model in a matter of minutes instead of hours.

#### 3.1 First results

With a significantly faster training time we have managed to get a a model that was able to achieve better results on the Lift task.



Figure 1: Training and validation loss as well as success rates across epochs of training a bare Decision Transformer on the Lift task.

One thing worth noticing is the variance of the success rates across epochs and the lack of correlation between the validation loss and the success rate. This is a problem raised by the robomimic paper. Because of this we have decided to evaluate 5 last models of the training of our models instead of just the best one (in terms of validation loss) - and to report the success rates across those 5 models.

We have also decided to evaluate the model - trained on only the red cube - on the blue and green cubes. It showed similar zero-shot performance as on the red cube.